RBB3023
Embedded and IOT Systems

May 2026

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Course Management

- **Synopsis**
  - This course explores practical application of microcontrollers and design of embedded systems for advanced applications. It also covers concept of real-time operating systems and their practical applications.
- **Year/Semester: Year 3, Semester 2**
- **Credit Hours: 3**
- **Pre-requisites:**
  - Structured Programming and Interfacing
  - Digital Electronics
  - Microprocessor
- **Hours**
  - 2 hours of lectures/week
  - 2 hours of laboratory sessions/week

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Learning Outcomes

- **At the end of this course you will be able to:**

- **CLO1**
  - Develop software solutions to solve specific problems (PO3)
- **CLO2**
  - Develop hardware design for embedded systems applications (PO3)
- **CLO3**
  - Implement real-time applications on embedded systems (PO5)

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Course content outline

- **Microcontroller Overview (2 hrs)**
  - Overall microcontroller functional Units
  - Microcontroller Architecture
  - Microcontroller memory organization
- **C Programming for the Microcontroller (4+4)**
  - Built-in C functions
  - Pre-processor directives
  - Program development tools

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- **Additional Microcontroller Hardware Interfacing (4+4)**
  - Analog-to-Digital Conversion (ADC)
  - Pulse Width Modulation
  - Interrupts
  - Timers and counters
- **Microcontroller Serial I/O (4+4)**
  - Interfacing with UART
  - RS232
  - Asynchronous
  - Synchronous
- **Real-Time Operating Systems (RTOS, 4+4)**
  - Tasks, semaphores, message queues
  - Exception and Interrupt handling
  - Scheduling
  - Memory management
  - Synchronization
  - Resource management

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Grading

- **Coursework:**
  - Practical Tasks: 20%
  - Tests: 30%
  - Quizzes / In Class Activity : 20%
  - Project: 30%
- **TOTAL: 100%**
- **Contact: patrick_sebastian@utp.edu.my**
- **Office : 23-03-21**
- **Consultation Hours:**

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Lab : Wednesday : 10 am – 12 pm :

23-02-07 (Microprocessor Lab)

Lecture : Monday (10am-12pm) – 23-02-10

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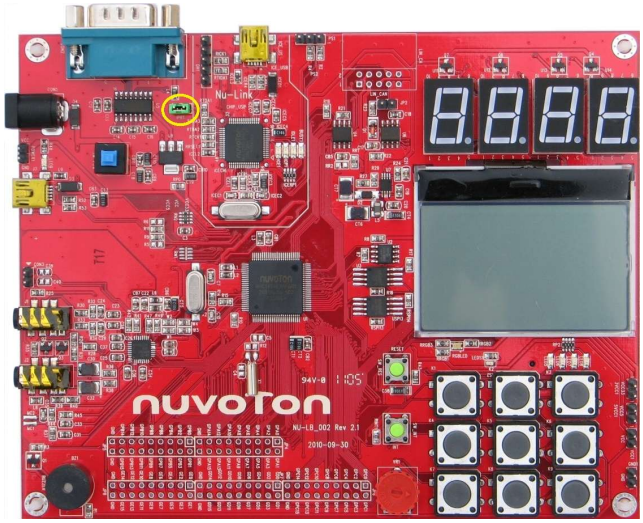
System development tools

- **Hardware**
  - Development board
    - **STM32 Nucleo-F411RE**
  - I/O devices
  - Breadboard

- **Software**
  - MKD5 – Keil compiler + uVision IDE
  - Mbed online IDE – cloud-based IDE
  - Mbed studio – Free to download and use
    - **Mbed OS 6**
    - **RTOS APIs**

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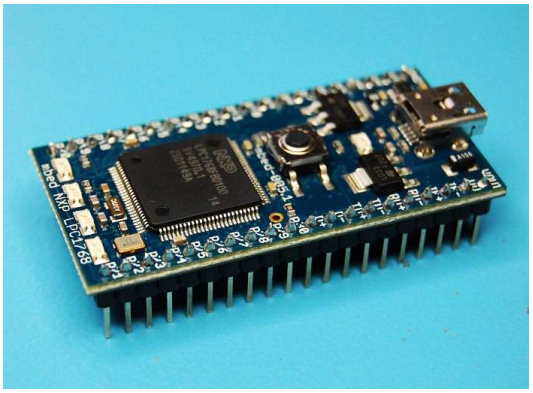
ARM Start-Up Kit [1]: NUVOTON LEARNING BOARD (Cortex M0+)



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ARM Start-Up Kit [2]: NXP mbed LPC1768 (Cortex M3)





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Nucleo-F411RE development board

- **STM32 Nucleo-64 development board**
  - with STM32F411RE MCU, supports Arduino and ST morpho connectivity
- **Features**
  - STM32 microcontroller in LQFP64 package
  - 1 user LED, 1 user and 1 reset push-buttons
  - 32.768 kHz crystal oscillator
  - Arduino™ Uno V3 expansion connector
  - ST morpho extension pin headers full access to all I/Os
  - On-board ST-LINK debugger/programmer
  - Arm® Mbed Enabled™ compliant



https://www.st.com/en/evaluation-tools/nucleo-f411re.html

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Types of software we'll develop

- **Bare metal (assembly language / C language)**
  - MDK5 (Keil compiler + uVision IDE)

- ~~**Bare metal (C/C++ language)**~~
  - ~~Mbed OS bare metal profile~~
  - ~~Subset of Mbed OS application programming interface (API)~~

- **RTOS-based**
  - Mbed OS full profile / MDK 5
  - Including use of RTOS APIs
  - Scheduler, threads, events, queue, semaphore

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